Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	22-10-2026
Team ID	
Project Name	i-Movies – Movie Ticket Booking System
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the i-Movies movie ticket booking system.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Accounts	Customer registration and login; separate administrator login; password hashing and token-based authentication; customer profile viewing and updates.
FR-2	Movie Discovery	Browse movies; search and filter the movie catalogue; view movie details.
FR-3	Theatre & Show Discovery	View theatres associated with movies; view showtimes and seat availability for a selected show.
FR-4	Seat Selection & Booking	Select available seats; create a booking; prevent booking seats that have already been reserved; handle booking creation failure without leaving seats incorrectly reserved.
FR-5	Booking Management	View personal bookings; cancel a booking; update booking status and release seats as part of cancellation.
FR-6	Favourites	Add movies to a customer's favourites and access the favourites list.
FR-7	Reviews & Ratings	Submit movie ratings and reviews.
FR-8	Administrator Content Management	Create, update, and delete movies, theatres, and shows; configure seat tiers and prices; view platform bookings; provision administrator accounts.
FR-9	User Experience Features	Use the theme preference and in-app notifications documented by the project.

Non-functional Requirements

The following non-functional requirements describe quality expectations for i-Movies. Where the repository does not specify a measurable target, the requirement is expressed without inventing one.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The platform should provide a clear, consistent interface for customers and administrators, making movie discovery, seat selection, booking, and content management straightforward.
NFR-2	Security	Passwords are hashed using bcryptjs. Authentication uses JWTs, and protected endpoints apply token verification and role-specific authorization. Production deployments should use HTTPS and restrict CORS to trusted frontend origins.
NFR-3	Reliability & Booking Integrity	Seat reservation should avoid double-booking by conditionally updating show seat availability. If booking creation fails, the reservation update should be rolled back; cancellation should update booking status before seats are released.
NFR-4	Performance	Movie discovery, showtime retrieval, and booking actions should respond efficiently. The repository does not specify a measured response-time target.
NFR-5	Availability & Deployment	The frontend is deployed separately from the backend. The frontend API base URL, backend database connection, JWT secret, and allowed CORS origins must be

NFR No.	Non-Functional Requirement	Description
		configured correctly for the deployed services to communicate.
NFR-6	Maintainability & Scalability	The application separates frontend UI, API routes/controllers/middleware, and Mongoose data models. This separation supports maintenance and future extension; a microservices architecture or a 99.9% uptime target is not documented.